

Answer Key — Biology Grade 11

Sub-Strand 3.3: Excretion and Homeostasis in Animals • Lesson 1: Same Hot Afternoon, Different Bodies: Launching the Excretion and Homeostasis Puzzle

Teacher reference only. Do not project or hand out.

Quick marking grid

Q1	Q2	Q3	Q4	Q5	Q6	Q7
C	C	A	D	B	C	C

USE AFTER: OBSERVE PHASE — Observe: group comparison table of excretory structures and nitrogenous wastes (freshwater fish row)

Q1. Learners compared tilapia from Lake Victoria with other animals in their group tables. What is the main nitrogenous waste of a freshwater fish such as tilapia?

- A. Freshwater fish do not make any nitrogenous waste
- B. Uric acid
- C. Ammonia**
- D. Urea

Correct answer: C. The charts show that freshwater fish excrete ammonia through the gills and in lots of dilute urine. Uric acid belongs to insects and birds, and fish do make nitrogenous waste.

USE AFTER: OBSERVE PHASE — Observe: group comparison table (insects row, excretory organ/structure column)

Q2. A locust and a housefly are insects. Which structures do they use to remove nitrogenous waste from the body?

- A. Kidneys only
- B. Lungs
- C. Malpighian tubules**
- D. Gills with chloride cells

Correct answer: C. The reference texts list Malpighian tubules as the excretory structure of insects, and they produce uric acid. Gills with chloride cells belong to marine fish.

USE AFTER: OBSERVE PHASE — Observe: group comparison table (birds row) and the chicken droppings example

Q3. The white, paste-like part of a chicken's droppings is mainly which substance?

- A. Uric acid**
- B. Urea
- C. Ammonia
- D. Undigested food (egested waste)

Correct answer: A. Birds excrete uric acid through the kidneys as a white semi-solid paste. It is an excretory waste, not undigested food, which is egested.

USE AFTER: OBSERVE PHASE — Observe: guiding questions 'Which animals need to save water...?' and 'Which animals have plenty of water...?'

Q4. Groups compared animals that must save water with animals that have plenty of water. Which pair of statements matches the pattern in their tables?

- A. Animals that must save water make ammonia in lots of dilute urine, and animals with plenty of water make uric acid paste
- B. All animals make the same waste and use the same amount of water
- C. Animals with plenty of water make uric acid, and animals that save water make ammonia
- D. Animals that must save water make uric acid paste using very little water, and animals with plenty of water make ammonia in dilute urine**

Correct answer: D. The tables link little water availability (insects, birds) to uric acid paste with very little water used, and plenty of water (freshwater fish) to ammonia in dilute urine.

USE AFTER: EXPLAIN PHASE — Explain: writing why freshwater fish excrete lots of dilute urine, using evidence from the tables

Q5. Which statement correctly compares the urine of freshwater fish and marine fish, as recorded in the group tables?

- A. Freshwater fish produce very little concentrated urine, while marine fish produce large amounts of dilute urine
- B. Freshwater fish produce large amounts of dilute urine, while marine fish produce very little concentrated urine**
- C. Both produce large amounts of dilute urine
- D. Neither freshwater fish nor marine fish produce urine

Correct answer: B. The charts show that freshwater fish such as tilapia excrete lots of dilute urine, while marine fish excrete very little concentrated urine. Fish do urinate.

USE AFTER: MODEL BUILDING PHASE — Model: first class model with question marks for the goat's kidney, dog's panting and lizard's basking

Q6. After this lesson the class has data on fish, insects and birds. Which part of the hot-afternoon puzzle is still NOT yet explained, and is marked with a question mark on the class model?

- A. Which nitrogenous waste freshwater fish such as tilapia make
- B. Why the white part of chicken droppings is a paste and not a liquid
- C. How a goat that has been drinking water produces concentrated urine**
- D. Which structures insects use to remove waste

Correct answer: C. The lesson names the goat's concentrated urine, the dog's panting and the lizard's basking as still unexplained. The fish, insect and bird waste data have already been studied.

USE AFTER: DQB PHASE — DQB: each learner writes 2 how/why questions on sticky notes about the phenomenon

Q7. Learners wrote questions on sticky notes for the Driving Question Board. Which question is the best 'how or why' question for investigating the goat in the puzzle?

- A. Is a goat an animal?
- B. What is urine?
- C. How do goats make concentrated urine even after they drink water?**
- D. Name one animal that lives in Lake Victoria.

Correct answer: C. A good Driving Question Board question starts with how or why and comes from part of the phenomenon that is not yet explained. The other choices are recall questions or do not come from the puzzle.